

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 14: Flight Mechanics, V-n Diagram & Asymmetric Flight

Syllabus Source: CAE Oxford (pp. 505-548) | Pass Target: $\geq 90\%$ AviationExam

1. COORDINATED TURNS: LOAD FACTOR, RADIUS & RATE

In a level coordinated turn, aerodynamic lift is tilted inward at bank angle Φ , providing the horizontal centripetal force ($F_c = L \times \sin(\Phi)$):

- Load Factor (n): Vertical equilibrium requires $L \times \cos(\Phi) = \text{Weight (W)}$. Therefore: Load Factor $n = \text{Lift} / \text{Weight} = 1 / \cos(\Phi)$. At 60° bank: $n = 1 / \cos(60^\circ) = 2.0 \text{ g}$!
- Turn Radius Formula: Radius (R) = $V^2 / (g \times \tan(\Phi))$. Turn radius depends EXCLUSIVELY on True Airspeed and Bank Angle! For a given bank angle, radius increases with speed squared (doubling TAS quadruples turn radius!).
- Rate of Turn (ROT): $\text{ROT} = (g \times \tan(\Phi)) / V$. Rate 1 Turn = $3^\circ/\text{second}$ (360° turn in 2 minutes). Rule of thumb: Bank Angle for Rate 1 = $(\text{TAS} / 10) + 7$ (e.g. at 140 kt: $14 + 7 = 21^\circ$ bank).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Maneuvering Speed VA & Load Factor

Golden Rule of VA in Exams: Step 1 (Below VA): Full abrupt control deflection will cause an AERODYNAMIC STALL before exceeding limit load factor ($+2.5\text{g}$ transport / $+3.8\text{g}$ normal). The wing stalls and relieves load, protecting structure. Step 2 (Above VA): The wing has sufficient lift available to exceed limit load factor before stalling. Full abrupt deflection will cause STRUCTURAL DAMAGE or catastrophic failure. Formula: $V_A = V_{s1} \times \sqrt{n_{\text{limit}}}$.

2. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Load Factor and Turn Stall Speed in a Steep Turn

SCENARIO (Fundamental Flight Mechanics Exam Drill): An aircraft has a level, unaccelerated 1g stall speed $V_s = 60$ knots: - The pilot executes a level coordinated steep turn at 60° bank angle ($\Phi = 60^\circ$) QUESTION: What is the load factor (n), and what is the new accelerated stall speed (V_{st}) in the turn? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Load Factor (n): - Formula: $n = 1 / \cos(\text{Bank Angle})$ - Bank Angle = $60^\circ \rightarrow \cos(60^\circ) = 0.50$ exactly! - Load Factor $n = 1 / 0.50 = 2.0 \text{ g}$! - (Both the aircraft structure and pilot experience twice their normal weight!). Step 2: Calculate the Accelerated Turn Stall Speed (V_{st}): - Master Formula: $V_{st} = V_s \times \sqrt{n}$ - $V_s = 60$ knots - $\sqrt{n} = \sqrt{2.0} = 1.414$ - $V_{st} = 60 \text{ kt} \times 1.414 = 84.8$ knots (~ 85 knots)! FINAL ANSWER: Load Factor = 2.0 g . Turn Stall Speed = 85 knots (a 41.4% increase in stall speed!). If airspeed drops below 85 kt during a 60° bank turn, the aircraft will stall immediately, even though it was flying safely above its 60 kt straight stall speed!

3. THE V-N DIAGRAM (MANEUVERING FLIGHT ENVELOPE)

The V-n diagram defines the structural and aerodynamic boundaries within which an aircraft may be safely operated:

Speed / Boundary on V-n Diagram	Regulatory Definition (CS-25 / CS-23)	Structural & Aerodynamic Limit Meaning
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Curved Left Boundary (Stall Line)	$n = C_{L_max} \times (1/2 \rho V^2) / (W/S)$.	Aerodynamic stall limit. Left of this line, the aircraft cannot generate enough lift to fly.
Maneuvering Speed (V_A)	Intersection of positive stall curve and limit load factor.	The MAXIMUM speed at which FULL, ABRUPT aerodynamic control deflection can be applied without exceeding structural limit load! (Aircraft stalls before structural failure).
Positive Limit Load Factor	+2.5 g (Transport CS-25); +3.8 g (Normal CS-23).	Maximum certified load factor without permanent structural deformation.
Ultimate Load Factor	1.5 x Limit Load Factor (+3.75 g for CS-25).	Airframe must withstand ultimate load for AT LEAST 3 SECONDS without catastrophic failure!
Design Dive Speed (V_D / M_D)	Maximum structural speed envelope limit.	Highest speed demonstrated in dive tests; margins above V _{MO} / M _{MO} .

4. MULTI-ENGINE ASYMMETRIC FLIGHT & V_{MCA}

When an engine fails on a multi-engine aircraft, asymmetric thrust creates severe yaw and roll toward the dead engine:

- The Critical Engine: On aircraft with conventional clockwise-rotating propellers, failure of the LEFT ENGINE is critical because the right engine's center of thrust acts farther from the fuselage centerline due to: 1. P-Factor (Asymmetric Blade Effect); 2. Accelerated Slipstream; 3. Torque Reaction; 4. Gyroscopic Precession.
- Minimum Control Speed Airborne (V_{MCA}): The minimum calibrated airspeed at which directional and lateral control can be safely maintained with the critical engine inoperative, maximum take-off power on the operating engine, and up to 5° OF BANK HELD INTO THE OPERATING ENGINE! If airspeed drops below V_{MCA}, rudder authority is lost and catastrophic uncontrollable roll/spin occurs!

[!] EXAM TRAP (AVIATIONEXAM) - V_{MCA} Bank Angle & Weight Dependencies

- 5° Bank Rule: Banking up to 5° TOWARDS THE LIVE ENGINE uses horizontal lift to assist the rudder, REDUCING V_{MCA} significantly! - Wings Level with Ball Centered: Increases V_{MCA} by 15-20 knots! Highly dangerous! - Weight Effect: Heavier aircraft has LOWER V_{MCA} (horizontal component of heavier lift vector helps counter yaw)! - Density Altitude: Higher altitude reduces engine thrust, which REDUCES yaw moment and LOWERS V_{MCA}!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the aerodynamic definition and safety significance of the 'Maneuvering Speed' (V_A)?

- [A]: The maximum speed for extending flaps
- [B]: The maximum speed at which full, abrupt deflection of flight controls can be made without risk of exceeding structural limit load factors, because the aircraft will stall aerodynamically before breaking
- [C]: The minimum drag speed

- [D]: The best rate of climb speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At speeds at or below V_A , the wing will stall before reaching the structural limit load factor (+2.5g on transport, +3.8g on utility), preventing structural overload during abrupt maneuvers.

Q2: Under CS-25 regulations, what is the 'Ultimate Load Factor' requirement for transport category aircraft?

- [A]: Exactly equal to the limit load
- [B]: 1.5 times the Limit Load Factor, which the structure must withstand for at least 3 seconds without collapsing
- [C]: 9.0 g
- [D]: 4.0 g

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Ultimate load incorporates a safety factor of 1.50 over the limit load (+2.5g x 1.5 = +3.75g). The airframe must sustain ultimate load for at least 3 seconds without structural collapse.

Q3: Why is the LEFT engine considered the 'critical engine' on conventional twin-engine piston aircraft with clockwise rotating propellers?

- [A]: Left engine has lower horsepower
- [B]: The right engine's center of thrust acts at a greater lateral moment arm from the fuselage centerline due to P-factor (asymmetric blade loading), generating more severe yaw if the left engine fails
- [C]: Left engine drives the alternator
- [D]: Propeller wash affects the cockpit

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Due to P-factor at high angles of attack, the descending blade produces more thrust. For clockwise propellers, this shifts the right engine's thrust line farther right, creating a larger yaw moment than the left engine.